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MyDesign Portfolio

Element E Initial Template - AT - 15 points

1 High Priority**# 2 High Priority****# 3 High Priority****# 4 Low Priority****# 5 Low Priority**

Element E: Application of STEM Principles and Practices

The design requirement for the prototype would have to be durable enough to repeatedly (a car driving back and forth five times over it) withstand the weight of a car when it drives over as well as water damage, expansion, and contraction due to temperature changes.

The STEM concepts involved in the requirement are physics, materials science, and civil engineering.

The physics apply directly to the specifics as we will have the gauge the effects of pressure, force, and friction on the patch material, materials science as we will have to understand the physical and chemical properties like rigidity and porosity of what we use as a patch material, and civil engineering as it involves the planning process, prototypes, and construction, and maintenance of our solution.

Dr. Sbar's notes:

- "If it penetrates through the patch and accumulates between patch and the base support, it can exert pressure on the patch especially during freezing and thawing."
- This project must be of limited scope, only focused on the patch not the whole patch and road structure. It seems better to focus on patch that is dense, robust and resists cracking.

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The asphalt would need to be a square yard road patch that should be run over for half an hour by a device that is 5% of the car's actual weight.

A scientific concept that is needed for the project is chemistry. Using chemistry, we can figure out which polymer is the best that can keep the gravel together the best and will plastically deform which will prevent cracks from forming. A technological requirement that would be necessary for the project is a block grid made of asphalt that can contract or expand in a way if there is water pressure underneath. An engineering concept that is needed for the design is modular construction. Modular construction can be used in the design by having a system of building blocks within the asphalt that will create a more sturdy surface that can resist overweight vehicles, as well as an adhesion substance that keeps the asphalt gravel together more effectively when dealing with high-speed vehicles and intense rainfall. A mathematical concept that would help in the process is a precise equal measurement of space between every single asphalt building block to not have rainwater blocked within the asphalt.

Engineering principles, such as masonry construction (the engineering behind block structures), are applied by designing interlocking asphalt blocks or segments that distribute weight evenly when vehicles drive over them. This reduces stress on any one part of the patch and increases overall durability. Mathematics can be used to have precise measurements of gravel.

Science will test out the flexibility of polymers and we can see how they react to their environment and see if they deform in a plastic manner to keep the road together. Technology can be used to make grip rebars flexible enough to keep asphalt together and prevent potholes. Potholes are caused by water seeping into cracks on the road, weakening the foundation. In cold climates, this water freezes and expands, causing the pavement to crack further.

Dr. Sbar's notes: So the mix consists of petrochemical polymer, gravel particles, and probably an adhesion promoter. Maybe some other stuff too. The adhesion promoter is critical. It ensures good chemical bonding between the polymer and the gravel stone surfaces. This gives strength, yet with the polymer, some plastic deformation is possible.



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The solution should be high quality in all aspects, such as making it durable, clean, effective, as well as balancing being easy & cheap to make, including having good material & longevity in use.

EDP: Engineering design process: involves repeated cycles of brainstorming, designing, building, testing, and analysis. This allows identification and resolution of flaws early in development, saving significant time and money later on. It also creates opportunities to improve designs.

DFM: Design for Manufacturability: designing the product/prototype with manufacturability as a requirement or major factor.

Modularity: Using modular variables in the product allows for cheap & long-lasting products, with components being able to be switched out (without needing repairs for the entire product), ensuring flexibility & longevity.

Material properties: Understanding material properties (strength, weight, heat resistance, conductivity, etc) helps in the selection of materials for specific performance requirements, like waterproofing, durability, cheapness, & especially cleanliness etc.

EDP - Engineering concept, which can be used as the template of creating a high-quality product.

DFM - Engineering concept, which can be applied when focusing on cheapness & easiness to produce.

Modularity - Design concept, can be applied when focusing on longevity, durability, & effectiveness.

Material properties - Science concept, can be applied when focusing on durability, effectiveness, longevity, & cleanliness.

Dr. Sbar's notes:

So the patch is applied hot. This allows the polymer to flow and become both mechanically integrated and chemically bonded to the surrounding hole materials. Pressure is applied to make the patch as dense as possible. So it would seem that a well bonded, dense, somewhat plastically deformable patch would contribute to robust pothole repair



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The design has to look aesthetically appropriate, like being the same color as the road instead of colors like bright red.

Science is used in this requirement since you need science to actually make the colors and pigments that we use to put on our project. The science concept that focuses on using color to attract attention is generally a combination of color psychology and color theory. One field of science used is chemical engineering as we need to determine the properties of whatever pigment we use to ensure that it can withstand the elements.

The texture of the asphalt used has to have enough friction/grip to ensure that tires don't slip.

The texture of the asphalt used has to have enough friction/grip to ensure that tires don't slip. Physics (friction and surface roughness), Materials Science (aggregate selection and bonding), and Engineering (surface testing and safety design).

Physics explains how friction between the tire and the road surface prevents slipping. By studying the coefficient of friction, we can determine the best texture and roughness for the asphalt. Materials science helps identify the right aggregate size, composition, and binder to achieve a durable, non-slippery surface even in wet conditions. Engineering applies these principles by testing various surface finishes and optimizing the asphalt mix for both grip and longevity.

Load distribution is the engineering and physics concept that explains how weight and forces are spread across a surface.

In our proposed design solution, this principle is used to make the asphalt patch durable enough to handle repeated pressure from vehicles driving over it. By evenly distributing the load through modular asphalt blocks and a reinforced base layer, we prevent



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concentrated stress points that could lead to cracking or deformation.

The design uses geometric patterns—such as interlocking or honeycomb structures—to disperse the car's weight evenly across the surface. This not only improves the patch's ability to handle heavy loads but also increases its lifespan and stability. Applying the concept of load distribution ensures that the patch can endure both constant movement and environmental changes, such as rain and temperature shifts, without losing structural integrity.



You can only work on the next sections after you have built and tested your first prototype. You need to skip to the "Evidence is Reviewed" section. You can skip over this section to get your STEM principles reviewed by an expert.

STEM Principles for Design Solution #2

STEM Concept: (Name a science, tech, engineering, or math idea you used to meet one of your top goals. "Ideas" list at the end of the assignment.)

We will be depending on the math idea of volume as our top goal involves conserving the volume of the prototype in consideration of environmental stresses like high/lower temperatures and water.

Describe how STEM concept applies directly to the specifics of your solution:

We have to gauge the volume of the prototype on every step of the testing phase to ensure that the volume stays constant within $\pm 5\%$ of the original volume.

STEM Principles for Design Solution #3 (if completed)



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STEM Concept: (Name a science, tech, engineering, or math idea you used to meet one of your top goals. "Ideas" list at the end of the assignment.)

Describe how STEM concept applies directly to the specifics of your solution:



Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder.

Documentation #1 (Only one documentation section needed for full credit):

What was the format you used to contact the expert: Interview notes, emails, audio/video tape with transcript, etc.

- What did the expert say about the connection you've made from your design requirements to STEM principles and practices:
- What did the expert say about how STEM principles and practices substantiate your design solution:



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Documentation #2 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What did the expert say about the connection you've made from your design requirements to STEM principles and practices:
- What did the expert say about how STEM principles and practices substantiate your design solution:

Documentation #3 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What did the expert say about the connection you've made from your design requirements to STEM principles and practices:
- What did the expert say about how STEM principles and practices substantiate your design solution:

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as experts. Use as many copies of this sub-template as needed. (Their job or hobby, why are they an expert?) You should have at least one expert.

Expert #1 (from documentation 1): Describe credential(s) and what makes them qualified as an expert.

Expert #2 (from documentation 2): Describe credential(s) and what makes them qualified as an expert.

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Expert #3 (from documentation 3): Describe credential(s) and what makes them qualified as an expert.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should tell you what you may want to change about your element E. Describe how those reviews informed your responses.

For example,

"_____ told us _____, which informed our decision to change/reconsider _____. "
(or "...which verified/reinforced our decision to _____.")

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will help you make iterations to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Interview with Expert #2 informs corrective response:

Interview with Expert #3 informs corrective response:

Use this table to help find your STEM Principle:

STEM Area	Concept Name	What It Means
Science	Newton's Laws	Explain how force, mass, and motion interact
Science	Conservation of Energy	Energy cannot be created or destroyed
Engineering	Design Process	Plan, build, test, and improve your idea
Engineering	Load Distribution	How weight and forces are spread out



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Math

Geometry

*Shapes, angles,
measurements*

Math

Area/Volume

*Space is something that takes
up*

Math

Data/Graphing

*Using data to support
decisions*

Technology

Coding Logic

*Rules and steps used in
programming*

Technology

Circuits

*Flow of electricity through
components*

Environmental Sci

Sustainability

*Using resources wisely,
reducing waste*

	Criteria	5	4	3	2	1	0
	3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>



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4.Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documented, relevant credentials</i>	<i>Two or more, with at least one with documented, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>
5.Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, but includes notes of attempts to gain feedback</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>